

3D aerodynamic summary

B12 wing with NASA_SC(2)-0712 sections. Generated by `python -m scripts.aero\_summary\_3d` from the results in wp2/results/; rerun it after the 3D or HLD analyses change. The same numbers, with their source files, are in aero_summary_3d.csv.

Configuration

quantity	value	unit	note
airfoil (root and tip)	NASA_SC(2)-0712		
S_{ref}	66.7	m ²	
AR	9.5	-	
taper λ	0.4	-	
span b	25.1724	m	
C_{root}	3.7853	m	
C_{tip}	1.5141	m	
MAC	2.812	m	
Λ_{LE}	31.9007	deg	
$\Lambda_{c/4}$	30	deg	
$\Lambda_{c/2}$	28.0236	deg	line of the 2D section reduction, Korn and the ADSEE Oswald formula
t/c (streamwise)	0.1199	-	
κ_A (Korn)	0.95	-	supercritical section
incidence i_w	2	deg	
tip twist	-2	deg	wash-out
dihedral Γ	2	deg	

Flight conditions

quantity	value	unit	note
cruise altitude	10668	m	
cruise M_∞	0.77	-	
cruise V_∞	228.3323	m/s	
$C_{L, design}$	0.4894	-	level-flight trim, WP1 weight and wing area
cruise M_n	0.6797	-	normal to the half-chord line
cruise $C_{l, n}$	0.6281	-	
cruise Re_n	1.325×10^7	-	on the normal MAC
landing V	65	m/s	
landing M_∞	0.191	-	sea level
landing Re_n	9.751×10^6	-	on the normal MAC

Cruise wing (VSPAERO + XFOil + Korn)

quantity	value	unit	note
α at $C_{L, design}$	-1.0493	deg	VSPAERO angle, the incidence comes on top
$C_{L\alpha}$	5.6928	1/rad	
$C_{L\alpha}$	0.09936	1/deg	
α_{0L}	-5.95	deg	
$C_{D, i}$	0.008245	-	lifting line on the VLM span loading
$C_{D, profile}$	0.007935	-	XFOil strips, sweep mode 'friction'

quantity	value	unit	note
$C_{D, wave}$	0.0003957	-	swept Korn + ADSEE drag rise
C_D	0.01658	-	all at CL_{design}
L/D	29.5271	-	at CL_{design}
L/D , no wave drag	30.2493	-	
L/D , $\cos^3 \Lambda$ sweep drag	34.7136	-	optimistic sensitivity
$(L/D)_{max}$	29.5513	-	
C_L at $(L/D)_{max}$	0.4944	-	
C_m (1/4 MAC)	-0.1611	-	at CL_{design} , inviscid
M_{dd}	0.8512	-	at CL_{design}
C_L at first section stall	0.9179	-	

Oswald factor and span efficiency

quantity	value	unit	note
e , ADSEE-II	0.5804	-	whole aircraft, all lift-dependent drag; from AR and sweep $c/2$ only
$K = 1/(\pi e AR)$, ADSEE-II	0.05773	-	
$C_{D,i}$ with ADSEE K	0.01383	-	at CL_{design}
$C_{D,i}$, lifting line	0.008245	-	wing only, inviscid, at CL_{design}
ϕ , lifting line	0.9735	-	from $\Gamma(y)$ on the wing; the CD_i used in the wing CD
$C_{D,i}$, VSPAERO near field	0.007995	-	wing only, inviscid, at CL_{design}
ϕ , VSPAERO near field	1.0039	-	$CD_{tot} - CD_0$, surface-integrated panel forces; comparison only
$C_{D,i}$, VSPAERO far field	0.009979	-	wing only, inviscid, at CL_{design}
ϕ , VSPAERO far field	0.8043	-	CD_{iw} , Trefftz plane; comparison only
effective e of the wing polar	0.9127	-	$CD = CD_0 + K CL^2$ over CL 0.2-0.8; wing with profile and wave drag, no fuselage
$C_{D,0}$ of the fit	0.007862	-	
K of the fit	0.03671	-	

The figures are different quantities, not competing estimates of one number.

- ADSEE-II e is the Oswald factor of the whole aircraft: all lift-dependent drag (induced, viscous, fuselage). The formula is the DATCOM lift-curve slope written through the lifting-line relation $CL_{\alpha} = 2\pi / (1 + 2/(AR e))$, so it reads the $\cos(\text{sweep})$ loss of lift slope as lost induced efficiency and drops steeply with sweep. It is the one for the aircraft drag polar $CD = CD_0 + K CL^2$.
- The span efficiencies are wing only and inviscid. The lifting-line value is the one in the wing CD . VSPAERO's near field lies just above 1 here and goes negative near $CL = 0$; its far field (CD_{iw}) changes with sweep at an unchanged span loading, which Munk's stagger theorem rules out for induced drag. Both are kept for comparison only.
- The effective e of the wing polar adds the lift-dependent profile and wave drag of the wing to the lifting-line induced drag, but no fuselage.

Clean wing at landing (VSPAERO + DATCOM)

quantity	value	unit	note
M	0.191	-	
section $C_{l, max, n}$	2.2735	-	XFoil landing polar
$C_{L, max}$, clean	1.5889	-	first section reaches $cl_{max, n}$
α at $C_{L, max}$	14.4963	deg	first section stall
$C_{L\alpha}$	0.07909	1/deg	
α_{0L}	-5.9531	deg	

quantity	value	unit	note
LE sharpness ΔY	2.97	% c	y_u(6% c) - y_u(0.15% c), streamwise section
$\Delta\alpha_{C_{L,max}}$	3.4	deg	DATCOM chart at the LE sweep
α_{stall}	17.536	deg	CL_max / CL_alpha + alpha_0L + dalpha_CLmax
required $C_{L,max}$, landing	2.49	-	
required $C_{L,max}$, take-off	2	-	

High-lift devices (HLD v2 trade-off)

quantity	value	unit	note
selected configuration	Fowler		equal weights
score	0.5652	-	
TE device outboard end η	0.75	-	
$C_{L,max}$, landing	2.4951	-	
$C_{L,max}$, take-off	2.2232	-	
α_{stall} , landing	14.8092	deg	

HLD v1: Pareto-optimal configurations

configuration	complexity	$\Delta C_{L,max}$	$C_{L,max}$ L	$C_{L,max}$ TO	α_{stall} L [deg]
triple-slotted + slat	7	1.5485	3.1374	2.6729	20.3262
triple-slotted + LE flap	6	1.461	3.0499	2.6116	21.5739
triple-slotted	5	1.2735	2.8624	2.4804	19.541
Fowler + LE flap	4	1.0937	2.6826	2.3545	16.7856
Fowler	3	0.9062	2.4951	2.2232	14.8092
single-slotted	2	0.6971	2.286	2.0769	17.4126
split	1	0.4826	2.0715	1.9267	14.7007
plain	1	0.4826	2.0715	1.9267	14.7007

L = landing, TO = take-off. Full table: hld/comparison.csv.

HLD v2: feasible configurations

#	configuration	cmplx	η_{TE}	$C_{L,max}$ L	$C_{L,max}$ TO	α_{stall} L	s perf	s space	s cmplx	score	Pareto
1	Fowler	3	0.75	2.4951	2.2232	14.8092	0.007955	0.6875	1	0.5652	*
2	triple-slotted	5	0.75	2.8569	2.4765	19.5199	0.5716	0.5	0.5	0.5239	*
3	Fowler + LE flap	4	0.75	2.6819	2.354	16.7779	0.2989	0.5	0.75	0.5163	*
4	double-slotted	4	0.75	2.6162	2.308	17.7237	0.1966	0.5938	0.75	0.5135	*
5	triple-slotted + LE flap	6	0.75	3.0437	2.6072	21.5448	0.8625	0.3125	0.25	0.475	*
6	double-slotted + LE flap	5	0.75	2.803	2.4388	19.808	0.4876	0.4062	0.5	0.4646	
7	Fowler + Krueger	5	0.75	2.6819	2.354	16.7779	0.2989	0.5	0.5	0.433	
8	triple-slotted + Krueger	7	0.75	3.0437	2.6072	21.5448	0.8625	0.3125	0	0.3917	
9	double-slotted + Krueger	6	0.75	2.803	2.4388	19.808	0.4876	0.4062	0.25	0.3813	
10	Fowler + slat	5	0.75	2.7701	2.4158	15.8716	0.4364	0.1875	0.5	0.3746	
11	single-slotted + slat	4	0.75	2.562	2.2701	18.4761	0.1122	0.25	0.75	0.3707	
12	triple-slotted + slat	7	0.75	3.1319	2.669	20.309	1	0	0	0.3333	*
13	double-slotted + slat	6	0.75	2.8912	2.5005	18.6522	0.6251	0.09375	0.25	0.3229	

L = landing, TO = take-off. Weights: performance 0.33, space 0.33, complexity 0.33. Weight sensitivity (66 weight sets): Fowler wins 39, triple-slotted + LE flap wins 12, triple-slotted + slat wins 10, triple-slotted wins 5. Full tables: hld_v2/tradeoff.csv, hld_v2/sensitivity.csv.